

BUILDING AN ANDROID IMAGE FROM SOURCE

Let us cover the different steps involved in building an Android image from source, and find out how to run it on a PC.



In just a few years, the Android OS has become a giant player in the global market. Now Android powers smartphones, televisions, tablets and a lot of other multimedia devices. Considering this, learning about Android will definitely be an advantage for a developer.

Why build Android from source? You may want a fully customised image for a specific hardware. You may also want to make Android images for different hardware. It is quite possible that you may not find an Android image for a specific hardware on the Internet. So, if you know how to build Android from source, you can make your own images.

But why run Android on your PC? The reasons could be many. If you don't have an Android smartphone to test the app you're developing, this could be an alternative. You might have an old PC that cannot run modern operating systems because they're too heavy, so you could run Android on it instead of throwing it away. Perhaps you have the urge to test-drive the latest Android release; or you might want to do so just because you are a geek!

Setting up the environment

I am using Ubuntu 11.10 on a 64-bit machine. If you are running a different OS or version, visit <http://source.android.com/source/initializing.html> to set up your environment. In general, you will need:

- Python 2.5—2.7, which can be downloaded from python.org
- GNU Make 3.81—3.82, which can be downloaded from gnu.org
- JDK 6 if you wish to build Gingerbread or a newer version; JDK 5 for Froyo or an older version.
- Git 1.7 or a newer version. You can find it at git-scm.com

Installing the JDK

Open the terminal and run the following commands:

```
sudo apt-get install python-software-properties
sudo add-apt-repository ppa:ferrnroberto/java
sudo apt-get update
```